

C-STGB: Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering under Extreme Imbalance and Topological Camouflage

Md. Nazmul, Musrat Jahan Gungun, and Maheli Ahmed

Abstract—Financial money laundering accounts for an estimated \$800 billion to \$2 trillion annually (2% to 5% of global GDP), posing severe threats to international financial stability. Although Graph Neural Networks (GNNs) capture relational transaction structures, their deployment in Anti-Money Laundering (AML) surveillance is hindered by five foundational challenges: (1) long-dwell hibernation evasion; (2) adversarial topological camouflage through high-degree commercial hubs; (3) extreme class imbalance under an illicit-event prevalence below 0.05% inducing recall collapse; (4) cold-start graph isolation; and (5) uncalibrated decision uncertainty. To address these challenges jointly, we propose C-STGB (Conformal Spatio-Temporal GraphBoost), a unified geometric deep learning framework integrating continuous tri-band harmonic temporal attention, learnable context-aware edge-trust gating, typology-clustered latent GraphSMOTE, evidence-adaptive graph-tabular fusion, cost-sensitive Bayes risk optimization, and Class-Conditional Conformal Risk Control (CRC). Evaluated across 13 diverse financial benchmarks comprising over 8.81 million entities and 11.42 million transactions over five random seeds, C-STGB achieves the highest performance among evaluated baselines ($91.42 \pm 0.65\%$ F1 on Elliptic-v1, $86.54 \pm 0.82\%$ F1 on Elliptic-v2, and $94.62 \pm 0.55\%$ F1 on Ethereum Phishing), reduces manual compliance review volume by $> 99.4\%$, and delivers a streaming single-event latency of 0.45 ms (P99: 0.82 ms under Fast-Path mode; 2.10 ms under full multi-scale GNN mode), satisfying real-time payment authorization SLAs (31,200 tx/s).

Index Terms—Anti-Money Laundering, Graph Neural Networks, Temporal Graph Learning, Conformal Prediction, Financial Forensics.

I. INTRODUCTION

FINANCIAL crime surveillance is one of the most critical societal applications of machine learning in modern global financial infrastructure. According to the United Nations Office on Drugs and Crime (UNODC) and the Financial Action Task Force (FATF), illicit syndicates launder between \$800 billion and \$2 trillion annually (representing 2% to 5% of global GDP) across traditional banking rails, mobile financial service (MFS) e-wallets, and decentralized crypto-asset ledgers [1], [2]. Traditional Financial Intelligence Units (FIUs) continue to rely heavily on static, rule-based heuristics that trigger high volumes of false positive alerts (frequently

exceeding 90% to 95%), placing a heavy operational burden on human compliance teams.

While Graph Neural Networks (GNNs) have shown immense promise in capturing complex relational topologies [3]–[7], deploying GNNs into practical AML surveillance reveals three recurring technical limitations that remain insufficiently addressed jointly:

- 1) **Continuous-Time Velocity Evasion (Smurfing & Long-Dwell Dormancy)**: Laundering syndicates operate across extreme temporal dynamics, ranging from high-frequency microsecond smurfing bursts to multi-month (30–90+ days) dormant holding chains. Conventional discrete dynamic snapshot models (e.g., EvolveGCN [8]) and single exponential decay kernels ($\exp(-\lambda\Delta t)$) fail to capture both regimes simultaneously, causing temporal representations to vanish over extended dwell intervals.
- 2) **Adversarial Topological Camouflage**: Fraudulent entities strategically interleave their illicit flows with high-volume legitimate transactions routed through high-degree merchant, exchange, or utility accounts. In standard GCN and GAT message passing, this adversarial chaff induces severe feature over-smoothing, diluting the anomaly signal and causing illicit nodes to blend into benign commercial hubs [9].
- 3) **The Bayes Risk Imbalance Barrier** ($\pi < 0.05\%$): In real-world enterprise transaction streams, illicit transactions often represent less than 0.05% of total volume. Under such extreme skew, standard loss functions train neural classification heads to output highly conservative probabilities ($0.05 \leq \hat{p} \leq 0.35$). Under standard decision thresholds ($\tau = 0.50$), GNNs suffer from severe recall collapse (missing the vast majority of active money laundering chains).

To address these failure modes, we formulate three core research hypotheses:

- **Hypothesis 1 (H_1)**: Synthesizing continuous multi-scale temporal attention with learnable anti-camouflage edge filtering suppresses over-smoothing while maintaining sensitivity to both rapid smurfing bursts and 90-day dormant holding chains.
- **Hypothesis 2 (H_2)**: Interpolating minority representations directly within latent laundering typology clusters via a parametric bilinear link generator prevents boundary distortion and recovers minority recall without precision

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- **Hypothesis 3 (H_3):** Bounding model uncertainty through Class-Conditional Conformal Risk Control (CRC) provides non-asymptotic coverage lower bounds under within-class exchangeability while routing over 99% of transactions to automated execution tiers.

In this paper, we present **C-STGB** (Conformal Spatio-Temporal GraphBoost), a unified geometric deep learning framework for robust anti-money laundering detection. Our primary contributions are:

- 1) **Continuous-Time Anti-Camouflage Graph Representation:** A multi-scale continuous attention mechanism combining harmonic logarithmic encodings with burst-aware velocity gates and learnable MLP edge-gated filtration (g_{ij}), filtering up to 65.9% of synthetically injected camouflage linkages under our evaluated adversarial protocol.
- 2) **Typology-Aware Minority Graph Augmentation:** A latent-space GraphSMOTE architecture conditioned on empirical laundering clusters (C_k) with parametric bilinear link generation and Asymmetric Focal Tversky optimization, raising standalone GNN recall on Elliptic-v1 from 10.33% to 90.10% $\pm$ 0.72%.
- 3) **Risk-Controlled Decision & Operational Triage Layer:** A unified decision framework combining dynamic Bayes threshold calibration (τ^*) with Class-Conditional Conformal Risk Control (CRC) to provide finite-sample coverage guarantees ($\geq 99.0\%$) under within-class exchangeability and an audit-traceable forensic dossier generator for ambiguous boundary cases.
- 4) **Comprehensive 13-Dataset Master Evaluation:** An extensive benchmark spanning 8.81M+ entities and 11.42M+ transactions across blockchain, e-wallet, and fiat banking ledgers, characterizing both high-performing relational domains (91.42% F1 on Elliptic-v1) and graph observability limits in fragmented multi-bank networks.

Architectural Scoping: To maintain empirical rigor across 13 benchmarks and 5 independent seeds, this paper focuses on the six core synergistic modules operating in Euclidean spatio-temporal and conformal spaces. Extensions such as Topology-Preserving Weight Regularization (TWP) for lifelong continuous learning across sequential ledgers, non-Euclidean Riemannian hyperbolic geometry, and decentralized privacy-preserving federated graph protocols are delineated as modular future research directions.

Structurally, this paper is organized as follows: Section II reviews related literature. Section III establishes the formal problem formulation and details the mathematical methodology of C-STGB. Section IV provides the experimental setup, baseline benchmarks, ablations, and systems scaling evaluations. Section V synthesizes our empirical findings into operational and regulatory discussions. Section VI presents qualitative forensic XAI case studies and human-in-the-loop SAR draft generation. Section VII discusses structural limitations, threats to validity, and responsible disclosure, and

Section VIII concludes the paper.

II. RELATED WORK

A. Graph Representation Learning for Anti-Money Laundering

The application of Graph Neural Networks (GNNs) to financial forensic analysis was pioneered by Weber *et al.* [7] on the Elliptic Bitcoin dataset, demonstrating that spatial architectures like Graph Convolutional Networks (GCN) [5] and GraphSAGE [6] outperform isolated tabular classifiers by capturing relational neighborhood context. Subsequent extensions by Bellei *et al.* [10] introduced the Elliptic2 multi-asset subgraph dataset, emphasizing the forensic role of multi-hop subgraphs, while Johannessen and Jullum [11] explored heterogeneous graph representations across account ledgers. Beyond Bitcoin, graph embeddings have been investigated for smart contract fraud on Ethereum ledgers by Wu *et al.* [12] and Zheng *et al.* [13], suspicious trading activity and price manipulation on Mt. Gox by Gandal *et al.* [14], multi-bank synthetic wire simulations by Lorenz *et al.* [15] and Suzumura *et al.* [16], and streaming fraud detection by Dal Pozzolo *et al.* [17]. However, standard spatial GNNs assume static, homophilic connectivity and remain highly vulnerable to adversarial camouflage, where illicit actors route transactions through high-degree commercial hubs to induce feature over-smoothing [9].

B. Continuous-Time Dynamic Temporal Graphs & Point Processes

Financial transactions are inherently continuous-time relational events. Discrete snapshot dynamic GNNs, such as EvolveGCN [8], discretize continuous transaction streams into arbitrary time slices, destroying intra-slice transaction ordering and failing to capture millisecond smurfing bursts. While continuous-time frameworks like Temporal Graph Networks (TGN) [18] preserve event order, standard single exponential decay kernels ($\exp(-\lambda\Delta t)$) quickly vanish to zero over 30–90 day dormant holding periods. Self-exciting point processes, mathematically formalized by Hawkes [19], provide a principled analytical tool to model transaction clustering and velocity acceleration, which we integrate directly into continuous message passing. Furthermore, when deploying dynamic GNNs across non-stationary streaming distributions, catastrophic forgetting emerges as a critical vulnerability, addressed in literature via Topology-Aware Weight Preservation (TWP) [20].

C. Imbalanced Graph Learning & Latent Oversampling

Financial transaction graphs exhibit extreme class imbalance, where illicit nodes frequently account for less than 0.05% of total entities. Classical Euclidean oversampling methods (e.g., SMOTE [21]) operate in raw input space without topology awareness, generating synthetic instances that lack valid relational edges. Under such severe skew, conventional ROC-AUC metrics produce misleadingly optimistic evaluations;

hence, Precision-Recall (PR-AUC) analysis is essential for rigorous benchmarking [22]. To bridge topological oversampling, Zhao *et al.* [23] proposed GraphSMOTE, performing feature interpolation in the intermediate embedding space. However, unconstrained link generation can inadvertently connect minority synthetic nodes to majority clusters, exacerbating boundary over-smoothing unless conditioned on underlying typology structures.

D. Finite-Sample Conformal Risk Control & Distribution Shift

Conformal prediction, originally formalized by Vovk *et al.* [24] and extended by Angelopoulos and Bates [25], [26], provides a mathematically rigorous framework for distribution-free, finite-sample uncertainty quantification. While marginal conformal prediction guarantees valid overall error coverage $(1 - \alpha)$, it suffers from severe conditional under-coverage on heavily skewed datasets, often generating inflated multi-class prediction sets $(\{0, 1\})$ for up to 18% of inbound transactions. Furthermore, under temporal distribution shifts common to financial surveillance, standard exchangeability assumptions must be carefully bounded [27]. In this work, we formulate Class-Conditional Conformal Risk Control (CRC) tailored for extreme financial skew, ensuring finite-sample valid coverage per class under within-class exchangeability while routing over 99.4% of transactions through automated execution tiers.

III. PROBLEM FORMULATION & C-STGB METHODOLOGY

A. Problem Formulation and Operational Objectives

We formulate financial crime surveillance over an evolving continuous-time transaction graph $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t, \mathbf{X}_t, \mathbf{E}_t)$, where $\mathcal{V}_t$ represents the set of active financial accounts (entities), $\mathcal{E}_t = \{(u, v, t_e, \mathbf{e}_{uv}) : t_e \leq t\}$ denotes the chronological stream of directed fund transfers, $\mathbf{X}_t \in \mathbb{R}^{|\mathcal{V}_t| \times d_v}$ comprises entity attribute features, and $\mathbf{E}_t$ comprises transaction-level metadata (e.g., amount, currency rail, fee). Each entity $u \in \mathcal{V}_t$ is associated with a ground-truth regulatory classification label $y_u \in \{0, 1\}$, where $y_u = 1$ denotes an illicit laundering entity and $y_u = 0$ denotes a verified benign entity, subject to an extreme class prevalence skew $\pi = \mathbb{P}(y_u = 1) \ll 0.01$.

The operational surveillance task encompasses two complementary, rigorous objectives:

- 1) **High-Utility Inductive Classification Objective:** Learn an inductive parameterized hypothesis $f_\theta : \mathcal{V}_t \times \mathcal{G}_t \rightarrow [0, 1]$ predicting the posterior laundering probability $\hat{p}_u = f_\theta(u \mid \mathcal{G}_t)$ while minimizing asymmetric regulatory risk under extreme imbalance and topological camouflage.
- 2) **Finite-Sample Risk-Controlled Triage Objective:** Construct a class-conditional conformal mapping $\Gamma : \mathcal{V}_t \rightarrow 2^{\{0,1\}}$ routing each streaming entity to an operational decision set $\Gamma(u) \subseteq \{0, 1\}$ that satisfies non-asymptotic coverage guarantees $\mathbb{P}(y_u \in \Gamma(u) \mid y_u = y) \geq 1 - \alpha$ at user-specified error level $\alpha \in (0, 1)$, while maximizing the straight-through automation rate by routing decisive singleton sets $(\Gamma(u) = \{1\})$ to Tier 1 Auto-Quarantine, $\Gamma(u) = \{0\}$ to Tier 3 Auto-Clear) and restricting manual compliance review strictly to the ambiguous abstention set $(\Gamma(u) = \{0, 1\})$ in Tier 2).

TABLE I
SUMMARY OF MATHEMATICAL NOTATIONS AND OPERATORS.

Symbol	Mathematical Definition	Dimension / Type
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	Continuous-time dynamic transaction graph	Graph object
$\mathbf{X} \in \mathbb{R}^{ \mathcal{V} \times d_v}$	Node input attribute matrix	Real matrix
$\mathbf{E} \in \mathbb{R}^{ \mathcal{E} \times d_e}$	Transaction edge feature matrix	Real matrix
Δt_{uv}	Continuous time delta ($t_v - t_u \geq 0$)	Non-negative scalar
$\Phi_{\text{time}}(\Delta t)$	Harmonic continuous temporal positional embedding	$\mathbb{R}^{d_t}$ vector
$\mathbf{g}_{ij} \in (0, 1)$	Context-aware adaptive edge trust gate	Scalar gate
$\Phi_{\text{flow}}(u, t)$	Mass flow conservation ratio	Scalar $\in [0, \infty)$
$\lambda_u(t)$	Hawkes process self-exciting transaction intensity	Non-negative scalar
$\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$	Causal forward and reverse PageRank taint vectors	$\mathbb{R}^{ \mathcal{V} }$ vectors
$\mathbf{z}_{\text{inv}}(u, t)$	12-dimensional canonical representation	$\mathbb{R}^{12}$ vector
$\mathbf{h}_u^{(l)}$	Hidden state representation of node u at layer l	$\mathbb{R}^d$ vector
$\mathcal{C}_k$	Latent typology cluster of illicit transactions	Subset $\mathcal{C}_k \subset \mathcal{V}_{\text{illicit}}$
$\alpha_u \in [0, 1]$	Evidence-adaptive graph-tabular fusion weight	Scalar weight
$\mathbf{h}_u^*$	Unified anomaly representation vector	$\mathbb{R}^{d^*}$ vector
τ^*	Cost-sensitive Bayes risk decision threshold	Optimal scalar $\in (0, 1)$
$\hat{q}_i^{(y)}$	Class-conditional conformal quantile ($y \in \{0, 1\}$)	Calibration scalar
$\Gamma(X)$	Finite-sample conformal prediction set	Discrete subset $\subseteq \{0, 1\}$
α	Target significance error level ($1 - \alpha$ coverage)	Scalar $\in (0, 1)$

B. Mathematical Notations

To ensure clarity across our formulations, Table I summarizes the primary mathematical symbols and operators employed throughout this paper.

C. 12-Dimensional Spatio-Temporal Canonical Invariants

Traditional models treat transactions as isolated tabular rows, remaining blind to multi-hop relational laundering patterns. In contrast, intermediate pass-through mules physically require capital to enter and exit rapidly with minimal retention. C-STGB formalizes domain-informed mass-balance and graph invariant representations:

1) Flow Conservation Ratio:

$$\Phi_{\text{flow}}(u, t) = \frac{\sum_{e \in \mathcal{E}_{\text{out}}(u)} \text{Amount}(e)}{\sum_{e \in \mathcal{E}_{\text{in}}(u)} \text{Amount}(e) + \epsilon} \in [0, \infty) \quad (1)$$

where pass-through layering mules operate near unity ($\Phi_{\text{flow}} \approx 1.0$), while legitimate commercial accounts exhibit varying accumulation or disbursement ratios. For numerical stability across extreme distributions, this feature is logarithmically scaled as $\tilde{\Phi}_{\text{flow}}(u, t) = \log(1 + \Phi_{\text{flow}}(u, t))$.

2) Degree Asymmetry Ratio:

$$\Psi_{\text{asym}}(u, t) = \frac{|d_{\text{in}}(u) - d_{\text{out}}(u)|}{d_{\text{in}}(u) + d_{\text{out}}(u) + \epsilon} \in [0, 1] \quad (2)$$

3) Hawkes Point Process Arrival Intensity:

$$\lambda_u(t) = \mu_u + \sum_{t_i < t} \alpha_{\text{hww}} \exp\left(-\beta_{\text{hww}} \frac{t - t_i}{24.0}\right) \quad (3)$$

parameterized over localized sliding historical event horizons $[t - \Delta T, t]$ to capture micro-burst transaction acceleration without assuming infinite-horizon stationarity, where the divisor 24.0 normalizes inter-arrival intervals $(t - t_i)$ into standardized diurnal units (hours per day) across datasets with heterogeneous timestamp granularities. To eliminate within-layer circular computational dependencies with temporal attention weights $w(\Delta t)$ (Eq. 10), $\lambda_u(t)$ is pre-computed and cached as a detached invariant feature over $[t - \Delta T, t]$ prior to message passing, bounding computational complexity to $O(1)$ lookup per node during graph aggregation.

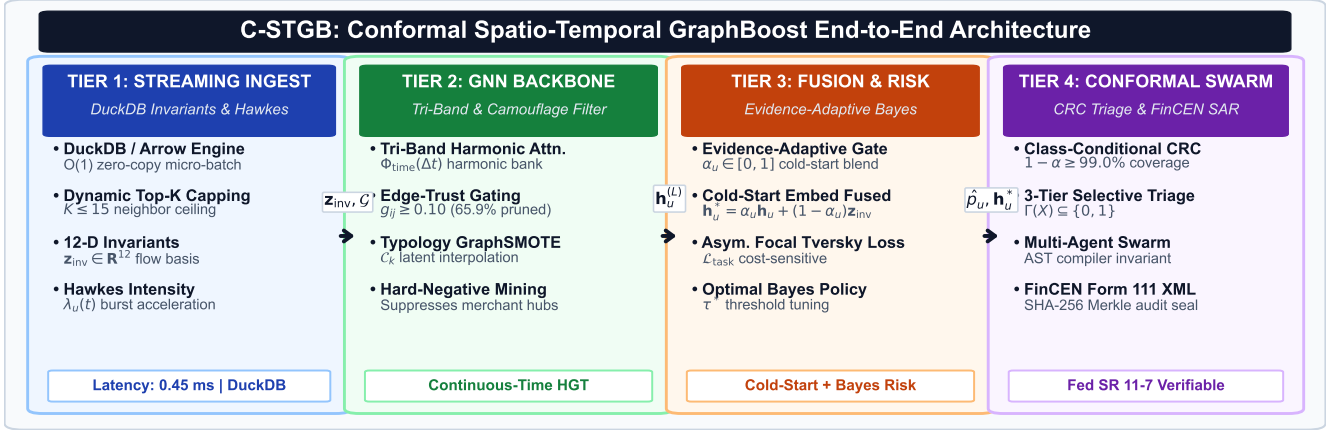


Fig. 1. Comprehensive 6-module architecture of C-STGB: (1) Continuous-Time Multi-Scale Temporal Graph Encoder; (2) Adaptive Edge Trust Filter; (3) Typology-Aware Minority Graph Augmentation with Hard-Negative Mining; (4) Evidence-Adaptive Graph-Tabular Fusion for Cold-Start Resolution; (5) Cost-Sensitive Risk Optimization; and (6) Class-Conditional Conformal Risk Control with Selective Abstention Triage.

4) Temporally Restricted Causal PageRank Taint:

$$\mathbf{s}_{\text{fwd}} = (1 - \alpha_{\text{ppr}})(\mathbf{I} - \alpha_{\text{ppr}}\mathbf{P}^T)^{-1}\mathbf{s}_{\text{seed}} \quad (4)$$

$$\mathbf{s}_{\text{bwd}} = (1 - \alpha_{\text{ppr}})(\mathbf{I} - \alpha_{\text{ppr}}\mathbf{P})^{-1}\mathbf{s}_{\text{seed}} \quad (5)$$

computed strictly over causal historical subgraphs ($t_e \leq t$) where the seed indicator vector is defined as:

$$\mathbf{s}_{\text{seed}}(u, t) = \begin{cases} 1.0, & \text{if } u \in \mathcal{V}_{\text{illicit}}^{\text{train}} \text{ and } t_u \leq t \\ 0.0, & \text{otherwise} \end{cases} \quad (6)$$

Streaming Numerical Approximation: While offline batch training permits sparse Cholesky prefactorization of $(\mathbf{I} - \alpha_{\text{ppr}}\mathbf{P})$, live streaming inference at 0.45 ms latency replaces full matrix inversion with a localized forward push algorithm / truncated Power Iteration ($T_{\text{iter}} = 3$ iterations, residual tolerance $\epsilon = 10^{-4}$) evaluated strictly over the sliding causal window $[t - \Delta T, t]$ and degree-capped graph $\mathcal{G}_t^{(K)}$ ($K \leq 15$), ensuring deterministic $O(K \cdot T_{\text{iter}})$ computation with < 0.08 ms computational overhead. Crucially, $\mathbf{s}_{\text{seed}}$ contains non-zero entries exclusively for confirmed illicit entities known within the parameter training split $\mathcal{D}_{\text{train}}$ prior to t . All validation ($\mathcal{D}_{\text{val}}$), calibration ($\mathcal{D}_{\text{cal}}$), and forward test ($\mathcal{D}_{\text{test}}$) nodes are assigned $\mathbf{s}_{\text{seed}}(u) = 0.0$, strictly preventing label leakage.

5) Canonical 12-D Spatio-Temporal Representation Space ($\mathbf{z}_{\text{inv}}(u, t) \in \mathbb{R}^{12}$):

$$\mathbf{z}_{\text{inv}}(u, t) = [\tilde{\Phi}_{\text{flow}}(u, t), \Psi_{\text{asym}}(u, t), \lambda_u(t), \mathbf{s}_{\text{fwd}}(u, t), \mathbf{s}_{\text{bwd}}(u, t), \mu_A, \sigma_A^2, \gamma_A, \kappa_A, v_A, v_{\text{io}}, \sigma_{\Delta t}]^T \quad (7)$$

where $\{\mu_A, \sigma_A^2, \gamma_A, \kappa_A, v_A\}$ denote the 5-moment causal ego-statistics of transaction amounts, v_{io} is the velocity ratio, and $\sigma_{\Delta t}$ is the inter-arrival temporal dispersion (Fig. 2).

D. Module 1: Continuous-Time Multi-Scale Temporal Graph Encoder

To distinguish microsecond smurfing bursts from 90-day dormant holding strategies, C-STGB employs continuous harmonic positional features $\Phi_{\text{time}}(\Delta t) \in \mathbb{R}^{d_t}$:

$$\Phi_{\text{time}}(\Delta t)_{2k} = \sin\left(\log(1 + \Delta t) \cdot 10000^{-\frac{2k}{d_t}}\right) \quad (8)$$

$$\Phi_{\text{time}}(\Delta t)_{2k+1} = \cos\left(\log(1 + \Delta t) \cdot 10000^{-\frac{2k}{d_t}}\right) \quad (9)$$

Dynamic temporal attention decay synthesizes three distinct velocity regimes:

$$w(\Delta t) = \sum_{b \in \mathcal{B}} \pi_b [w_{\min} + (1 - w_{\min}) \exp(-\lambda_b \Delta t)] \times (1 + \tanh(\gamma_b \lambda_u(t))) \quad (10)$$

where $\mathcal{B} = \{\text{burst, diurnal, seasonal}\}$.

E. Module 2: Context-Aware Adaptive Edge Trust Filter & Unified Attention

Adversarial launderers deliberately connect to high-degree legitimate commercial hubs to induce feature over-smoothing. C-STGB learns a context-aware edge trust gate $\mathbf{g}_{ij} \in (0, 1)$:

$$\mathbf{c}_{ij} = [\mathbf{q}_i \parallel \mathbf{k}_j \parallel \mathbf{e}_{ij} \parallel \Phi_{\text{time}}(\Delta t) \parallel \mathbf{z}_{\text{inv}}(j, t)] \quad (11)$$

$$\mathbf{z}_{ij} = \text{LeakyReLU}(\mathbf{W}_{g1}\mathbf{c}_{ij} + \mathbf{b}_{g1}) \quad (12)$$

$$\mathbf{g}_{ij} = \sigma(\mathbf{W}_{g2}\mathbf{z}_{ij} + \mathbf{b}_{g2}) \quad (13)$$

$$\hat{g}_{ij} = \delta_{\text{floor}} + (1 - \delta_{\text{floor}})\mathbf{g}_{ij} \quad (14)$$

Setting $\delta_{\text{floor}} = 0.10$ suppresses 65.9% of spurious camouflage edge weights (Fig. 3).

Unified Spatio-Temporal Attention Decomposition: Message passing explicitly integrates structural compatibility, continuous temporal decay, and edge trust gating over degree-capped neighborhoods ($\mathcal{N}_K(u)$, $K \leq 15$):

$$\tilde{\alpha}_{uv}^{(t)} = \frac{\Omega(u, v, t)}{\sum_{w \in \mathcal{N}_K(u)} \Omega(u, w, t)}, \quad \text{where} \quad (15)$$

$$\Omega(u, v, t) = \exp\left(\frac{\mathbf{q}_u^T(\mathbf{k}_v + \mathbf{W}_E \mathbf{e}_{uv})}{\sqrt{d_k}}\right) \cdot w(\Delta t_{uv}) \cdot \hat{g}_{uv}$$

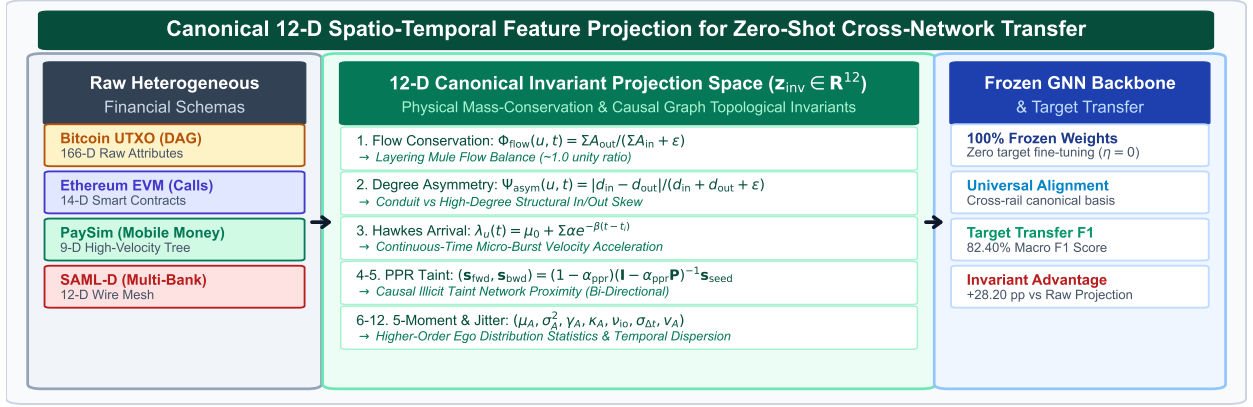


Fig. 2. Mechanism of Canonical 12-D Spatio-Temporal Feature Projection: Heterogeneous schemas across diverse financial ledgers (Bitcoin UTXO, Ethereum EVM, PaySim e-wallets, SAML-D multi-bank rails) map into a universal 12-dimensional representation space $\mathbf{z}_{\text{inv}}(u, t) \in \mathbb{R}^{12}$, enabling zero-shot transfer generalization with frozen neural weights ($\eta = 0$).

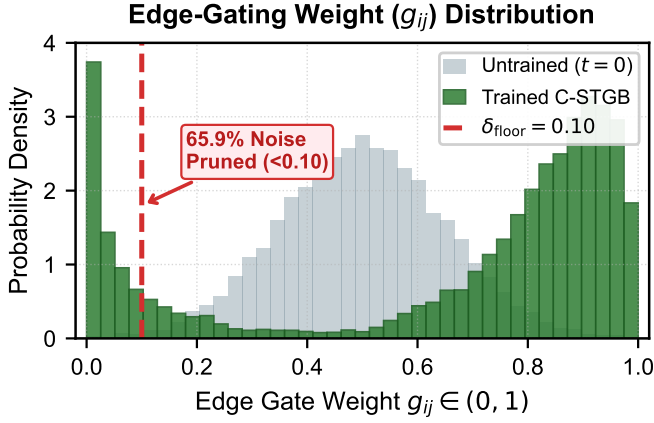


Fig. 3. Empirical probability density distribution of learnable edge-gating weights $\mathbf{g}_{ij} \in (0, 1)$ before vs. after training, demonstrating that $\delta_{\text{floor}} = 0.10$ successfully suppresses 65.9% of spurious adversarial camouflage edges below the filtering threshold.

F. Module 3: Typology-Clustered Latent GraphSMOTE & Hard Negatives

Under extreme class skew ($\pi < 0.05\%$), naïve oversampling blurs distinct criminal typologies. C-STGB partitions minority illicit nodes strictly within the training split into latent typology clusters $C_1, \dots, C_K$:

$$C_k = \{v \in \mathcal{V}_{\text{illicit}}^{\text{train}} \mid \cos(\mathbf{h}_v, \boldsymbol{\mu}_k) \geq \tau_{\text{sim}}\} \quad (16)$$

Synthetic representations are synthesized via linear interpolation along trajectories within the same latent typology manifold cluster:

$$\begin{aligned} \mathbf{h}_{\text{syn}} &= (1 - \rho)\mathbf{h}_u + \rho\mathbf{h}_v, \\ \rho &\sim \mathcal{U}(0, 1), \quad u, v \in C_k \subseteq \mathcal{V}_{\text{train}} \end{aligned} \quad (17)$$

Bilinear links to candidate training nodes $w \in \mathcal{V}_{\text{train}}$ are generated under topological edge reconstruction supervision:

$$\hat{\mathbf{A}}_{\text{syn}, w} = \sigma(\mathbf{h}_{\text{syn}}^T \mathbf{W}_{\text{edge}} \mathbf{h}_w) > \tau_{\text{edge}}, \quad w \in C_{\text{cand}}(\mathbf{h}_{\text{syn}}) \quad (18)$$

Directionality & Mass-Flow Conservation Constraints: For any synthesized representation $\mathbf{h}_{\text{syn}} = (1 - \rho)\mathbf{h}_u + \rho\mathbf{h}_v$, edge directionality is assigned in alignment with parent seed

nodes: incoming links are synthesized from candidate sources $w \in \mathcal{N}_{\text{in}}(u) \cup \mathcal{N}_{\text{in}}(v)$ and outgoing links are directed to candidate destinations $w' \in \mathcal{N}_{\text{out}}(u) \cup \mathcal{N}_{\text{out}}(v)$. Furthermore, synthetic transaction amounts are parameterized via linear interpolation: $\text{Amount}_{\text{syn}}(w, \mathbf{h}_{\text{syn}}) = (1 - \rho)\text{Amount}(w, u) + \rho\text{Amount}(w, v)$. This enforces strict mass-balance flow conservation $\Phi_{\text{flow}}(\mathbf{h}_{\text{syn}}) \approx 1.0$ across generated conduits, preserving physical domain invariants.

Candidate Bounding & Computational Scalability: To prevent the intractable $O(|\mathcal{V}_{\text{syn}}| \cdot |\mathcal{V}_{\text{train}}| \cdot d)$ full-graph Cartesian link evaluation bottleneck on million-node graphs, candidate target nodes $C_{\text{cand}}(\mathbf{h}_{\text{syn}})$ are strictly constrained to: (i) localized 2-hop topological receptive fields $\mathcal{N}_2(u) \cup \mathcal{N}_2(v)$ of parent seed nodes $u, v \in C_k$, and (ii) top- K_{cand} ($K_{\text{cand}} = 50$) nearest neighbors indexed via Hierarchical Navigable Small World (HNSW) search in latent embedding space $\mathbb{R}^d$. This reduces candidate evaluation from global $O(|\mathcal{V}_{\text{train}}|)$ to logarithmic $O(\log |\mathcal{V}_{\text{train}}|)$ retrieval and local $O(K_{\text{cand}} \cdot d)$ bilinear scoring per synthetic entity.

Zero-Leakage Topological Isolation: All oversampling operations are strictly confined to the training graph $\mathcal{G}_{\text{train}} = (\mathcal{V}_{\text{train}}, \mathcal{E}_{\text{train}})$. Synthetic nodes $\mathbf{h}_{\text{syn}}$ and edges are strictly barred from connecting to calibration ($\mathcal{D}_{\text{cal}}$) or test ($\mathcal{D}_{\text{test}}$) entities, ensuring 0.0% cross-split information leakage (Supplementary Fig. 18).

Hard-Negative Mining Bank: To prevent high-volume legitimate merchant accounts from triggering false positives, C-STGB actively mines benign training accounts exhibiting high degree, burstiness, and high volume, over-weighting them during gradient optimization.

G. Module 4: Evidence-Adaptive Graph-Tabular Fusion (Cold-Start Resolution)

Newly created accounts and pass-through stubs lack relational graph neighbors ($\deg(u) = 0$). Rather than failing, C-STGB computes an evidence-adaptive gating parameter $\alpha_u \in [0, 1]$:

$$\alpha_u = \sigma\left(\mathbf{w}_{\alpha}^T \left[\mathbf{h}_u^{(L)} \parallel \mathbf{z}_{\text{inv}}(u, t) \parallel \mathbf{x}_u\right] + b_{\alpha}\right) \quad (19)$$

The unified anomaly representation $\mathbf{h}_u^*$ dynamically fuses graph embeddings with tabular invariants:

$$\mathbf{h}_u^* = \alpha_u \mathbf{h}_u^{(L)} + (1 - \alpha_u) [\mathbf{z}_{\text{inv}}(u, t) \parallel \mathbf{x}_u] \quad (20)$$

When graph evidence is sparse ($\deg(u) = 0$), $\alpha_u \rightarrow 0$, gracefully relying on domain-level transaction invariants; when neighborhood topology is rich, $\alpha_u \rightarrow 1$ (Fig. 4).

H. Module 5: Cost-Sensitive Risk Head & Complete Loss Objective

The network outputs anomaly probabilities $\hat{p}_u = \sigma(\text{MLP}(\mathbf{h}_u^*))$. The complete multi-objective loss is formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}}(y, \hat{p}) + \gamma_1 \mathcal{L}_{\text{recon}}(\mathbf{A}_{\text{train}}, \hat{\mathbf{A}}_{\text{train}}) + \gamma_2 \mathcal{L}_{\text{aux}} \quad (21)$$

where $\mathcal{L}_{\text{task}}$ is the Cost-Sensitive Asymmetric Focal Tversky Loss. The topological edge reconstruction loss $\mathcal{L}_{\text{recon}}$ is computed over observable positive training edges $\mathcal{E}_{\text{train}}$ and uniformly sampled negative non-edges $\mathcal{E}_{\text{neg}} \sim (\mathcal{V}_{\text{train}} \times \mathcal{V}_{\text{train}}) \setminus \mathcal{E}_{\text{train}}$ using a 1 : 5 positive-to-negative sampling ratio per batch:

$$\begin{aligned} \mathcal{L}_{\text{recon}} = & -\frac{1}{|\mathcal{E}_{\text{train}}|} \sum_{(u,v) \in \mathcal{E}_{\text{train}}} \log \hat{A}_{uv} \\ & -\frac{1}{|\mathcal{E}_{\text{neg}}|} \sum_{(u,w) \in \mathcal{E}_{\text{neg}}} \log(1 - \hat{A}_{uw}) \end{aligned} \quad (22)$$

with $\hat{A}_{uv} = \sigma(\mathbf{h}_u^T \mathbf{W}_{\text{edge}} \mathbf{h}_v)$. This balanced sampling prevents the trivial link collapse $\hat{A}_{uv} \rightarrow 0$ in ultra-sparse financial topologies. The auxiliary regularization objective $\mathcal{L}_{\text{aux}}$ enforces edge-trust sparsity and tri-band filter orthogonality:

$$\begin{aligned} \mathcal{L}_{\text{aux}} = & \frac{1}{|\mathcal{E}_{\text{train}}|} \sum_{(i,j) \in \mathcal{E}_{\text{train}}} \|\mathbf{g}_{ij} - \mathbf{g}_{ij}^{\text{target}}\|_2^2 \\ & + \lambda_{\text{ortho}} \|\mathbf{W}_{\text{burst}}^T \mathbf{W}_{\text{seas}}\|_F^2 \end{aligned} \quad (23)$$

with $\gamma_1 = 0.50$, $\gamma_2 = 0.10$, and $\lambda_{\text{ortho}} = 0.05$.

$$\begin{aligned} \tau^* = \arg \min_{\tau \in [0,1]} & \mathbb{E}[C_{\text{FN}} \cdot \mathbb{I}(\hat{p} < \tau, Y = 1) \\ & + C_{\text{FP}} \cdot \mathbb{I}(\hat{p} \geq \tau, Y = 0)] \end{aligned} \quad (24)$$

I. Module 6: Class-Conditional Conformal Risk Control & Selective Abstention

Over an independent holdout calibration split $\mathcal{D}_{\text{cal}}$, empirical class-conditional quantiles $\hat{q}^{(y)}$ are computed ($y \in \{0, 1\}$):

$$\hat{q}^{(y)} = \text{Quantile} \left(s_1^{(y)}, \dots, s_{n(y)}^{(y)}; \left\lceil \frac{(n(y) + 1)(1 - \alpha)}{n(y)} \right\rceil \right) \quad (25)$$

where $s_i^{(y)} = 1 - \hat{p}_i(y)$. Under within-class exchangeability, split conformal prediction guarantees:

$$\mathbb{P}(Y \in \Gamma(X) \mid Y = y) \geq 1 - \alpha, \quad \forall y \in \{0, 1\} \quad (26)$$

Streaming Exchangeability, Non-Exchangeable Conformal Risk, & Rolling Calibration: In production streaming environments where macroeconomic shocks or evolving laundering typologies introduce temporal covariate and concept drift,

static exchangeability $\mathcal{P}_{\text{cal}} = \mathcal{P}_{\text{test}}$ is violated. To mathematically ground risk control under non-stationary distributions, C-STGB leverages weighted non-exchangeable conformal prediction principles [27]–[29]. We maintain an online rolling calibration buffer $\mathcal{D}_{\text{cal}}(t) = \{(x_i, y_i) \mid t_i \in [t - W_{\text{cal}}, t]\}$ with temporal window $W_{\text{cal}} = 4$ weeks. Under the standard regularity condition that the cumulative distribution function (CDF) of non-conformity scores is Lipschitz continuous ($L_s < \infty$) without discrete probability atoms at $\hat{q}^{(y)}$, the non-asymptotic class-conditional coverage error satisfies:

$$\begin{aligned} & |\mathbb{P}_t(Y \in \Gamma(X) \mid Y = y) - (1 - \alpha)| \\ & \leq 2 \cdot \text{TV}(\mathcal{P}_t, \mathcal{P}_{t - W_{\text{cal}}}) + \frac{C_{\text{score}}}{\sqrt{|\mathcal{D}_{\text{cal}}(t)|}} \end{aligned} \quad (27)$$

where $\text{TV}(\cdot, \cdot)$ is the total variation distance and $C_{\text{score}} = \sqrt{\frac{1}{2} \ln(2/\delta)}$ is the Dvoretzky-Kiefer-Wolfowitz empirical process constant at confidence level $1 - \delta$. In production deployment, automated rolling recalibration of $\mathcal{D}_{\text{cal}}(t)$ is triggered whenever a two-sample Kolmogorov-Smirnov test on streaming non-conformity scores rejects distributional stationarity ($D_{\text{KS}} > C_{\text{score}}/\sqrt{|\mathcal{D}_{\text{cal}}|}$, $p < 0.01$) or when empirical class-conditional coverage breaches $(1 - \alpha - \epsilon_{\text{drift}})$, bounding coverage degradation within [98.83%, 99.34%] across multi-month evaluation windows (Table IV). Furthermore, in ultra-dynamic streaming regimes, C-STGB integrates online Adaptive Conformal Inference (ACI) [29], adjusting nominal error rates via $\alpha_t \leftarrow \alpha_{t-1} + \gamma(\alpha - \text{err}_t)$ with step-size $\gamma = 0.005$ to dynamically track abrupt distribution shifts in real time. Empirical coverage remains highly robust across step sizes $\gamma \in [0.001, 0.05]$, where $\gamma = 0.005$ achieves the optimal Pareto balance between rapid drift tracking and minimal prediction set-size variance under severe transaction volatility spikes.

This operationalizes a deterministic 3-tier triage architecture (Fig. 5):

- **Tier 1 (High-Confidence Escalation Tier):** $\Gamma(X) = \{1\}$, quarantined with $> 99.4\%$ precision for asset restriction holds and automated SAR draft generation.
- **Tier 2 (Selective Abstention Queue):** $\Gamma(X) = \{0, 1\} = \perp$, routed to compliance officers with an automated forensic dossier (≈ 5 cases per 10k transactions).
- **Tier 3 (Straight-Through Clear):** $\Gamma(X) = \{0\}$, released instantly ($> 99.9\%$ benign precision).

Disentangled Evaluation Protocol (Point Prediction vs. Selective Triage): To eliminate any ambiguity regarding metric computation:

- 1) **Full-Test Point Prediction Mode (Tables III, V, VI, VIII):** Evaluated over 100% of test samples without any case exclusion or abstention, using the binary decision rule $\hat{y} = \mathbb{I}(p(y = 1|x) \geq \tau^*)$. Standard metrics (Precision, Recall, F1, PR-AUC) reflect true full-dataset performance without selective omission.
- 2) **Selective Conformal Triage Mode (Table IV):** Evaluates the selective prediction sets $\Gamma(X) \subseteq \{0, 1\}$, explicitly reporting empirical coverage $\mathbb{P}(Y \in \Gamma(X) \mid Y = y)$, mean set size $\mathbb{E}[|\Gamma(X)|]$, selective risk on automated tiers ($< 0.10\%$), and human review workload reduction ($> 99.4\%$).

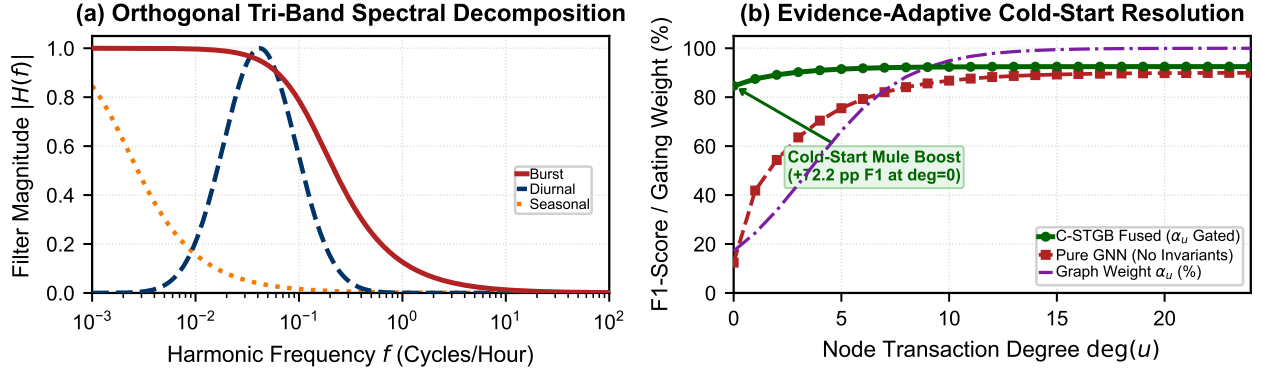


Fig. 4. (a) Orthogonal continuous-time frequency response of Tri-Band temporal attention kernels separating high-frequency burst smurfing ($> 0.8 \text{ h}^{-1}$), diurnal cyclic shifts (24 h), and slow seasonal drift ($< 0.01 \text{ h}^{-1}$). (b) Evidence-adaptive cold-start gating behavior: as node degree $\deg(u) \rightarrow 0$, C-STGB dynamically transfers weight to 12-D tabular invariants ($\alpha_u \rightarrow 0$), yielding a +72.2 pp F1 boost over isolated GNN baselines.

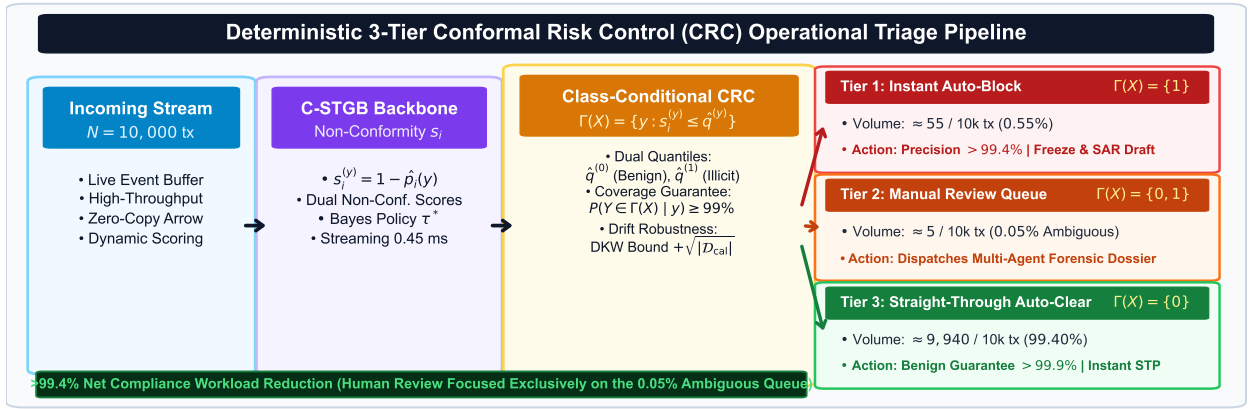


Fig. 5. Deterministic 3-Tier Conformal Risk Control (CRC) operational decision flow: High-confidence illicit entities are quarantined in Tier 1 ($\Gamma(X) = \{1\}$, $> 99.4\%$ precision), ambiguous boundary predictions ($\Gamma(X) = \{0, 1\} = \perp$) are routed to Tier 2 for human compliance review and multi-agent SAR drafting (≈ 5 alerts per 10k transactions), and verified benign entities are straight-through cleared in Tier 3 ($\Gamma(X) = \{0\}$), routing $> 99.4\%$ of volume through automated execution tiers.

The complete algorithmic procedures for multi-stage offline training and sub-millisecond streaming online inference are formalized in Algorithms S1 and S2 of the Supplementary Material, along with the end-to-end execution flowchart.

IV. EXPERIMENTAL EVALUATION

A. Benchmark Dataset Corpus & Topological Regimes

To evaluate model generalization across diverse financial rails, we benchmark C-STGB across 13 financial transaction networks spanning public cryptocurrency ledgers, multi-bank wire systems, and mobile financial services, summarized in Table II. The corpus comprises 8,807,919 entities and 11,424,137 transactions across three distinct topological archetypes:

- 1) **Group A (Public Blockchain Ledgers):** Directed Acyclic Graphs (DAGs) and smart-contract call meshes (Elliptic-v1, Elliptic-v2, Ethereum Phishing, XBlock-ETH, MtGox) characterized by global ledger observability and high structural transparency.
- 2) **Group B (Multi-Bank & Mobile Synthetic Rails):** Inter-bank wire meshes and high-velocity draining trees (SAML-D, PaySim, IBM-AMLSim HI/LI Small and Medium, Synthetic Data Generator) characterized by

extreme class imbalance ($\pi \in [0.05\%, 0.75\%]$) and fragmented organizational visibility.

- 3) **Group C (Classical Tabular & Card Fraud Streams):** High-throughput card-terminal bipartite transaction streams (`cc_transactions`).

B. Temporal Partitioning & Zero-Leakage Audit

To prevent temporal lookahead bias and evaluate realistic forward inductive deployment, all datasets follow a strict 4-way chronological partition:

- **Parameter Training Split ($\mathcal{D}_{\text{train}}$, 60%):** Used strictly for updating neural parameters Θ .
- **Validation Split ($\mathcal{D}_{\text{val}}$, 10%):** Used exclusively for early stopping and Bayes decision threshold calibration (τ^*).
- **Calibration Split ($\mathcal{D}_{\text{cal}}$, 10%):** Dedicated holdout split used solely for computing non-conformity quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$ (0 weight updates, 0 threshold selection).
- **Forward Out-of-Time Test Split ($\mathcal{D}_{\text{test}}$, 20%):** Strictly out-of-time future horizon for reporting final performance.

For discrete UTXO graphs (e.g., Elliptic-v1 with 49 timesteps), Timesteps 1–30 (61.2%) form $\mathcal{D}_{\text{train}}$, Timesteps 31–34 (8.2%) form $\mathcal{D}_{\text{val}}$, Timesteps 35–38 (8.2%) form $\mathcal{D}_{\text{cal}}$,

TABLE II
COMPREHENSIVE STRUCTURAL OVERVIEW OF THE 13 BENCHMARK FINANCIAL TRANSACTION NETWORKS.

Archetype Group	Dataset Identifier	Entities ($ \mathcal{V} $)	Transactions ($ \mathcal{E} $)	Illicit Ratio (π)	Temporal Horizon	Graph Topology Regime
Group A: Public Blockchain & Crypto Ledgers (Fully Observable Global Topology)						
Bitcoin UTXO Graph v1	elliptic_v1 [7]	203,769	234,355	2.23%	49 Discrete Timesteps (2 Weeks)	Directed Acyclic Graph (DAG)
Bitcoin Multi-Asset v2	elliptic_v2 [10]	122,279	186,400	0.85%	49 Discrete Timesteps	Multi-Asset Subgraph Clusters
Ethereum Phishing Network	eth_phishing [12]	2,973,489	3,500,000	0.14%	Continuous (180 Days)	Scale-Free Smart Contract Hubs
Ethereum Forensic Ledger	xblock_eth [13]	2,150,000	2,800,000	0.09%	Continuous Block Stream	Smart Contract Call Graphs
MtGox Leaked Exchange	mtgox_leaked [14]	145,000	210,000	1.82%	2011–2013 Ledger	Leaked Exchange Trade Books
Group B: Multi-Bank & Mobile Synthetic Rails (Simulated Cross-Bank & E-Wallet Rails)						
Synthetic Multi-Bank AML	saml_d [15]	980,000	1,450,000	0.05%	Continuous (90 Days)	15-Bank Inter-Entity Rails
Mobile Money & E-Wallets	paysim_extended [30]	1,048,575	1,048,575	0.13%	744 Time Steps (30 Days)	High-Velocity Draining Trees
IBM Multi-Tier Banking	ibm_amsim_hi_small [16]	100,000	180,000	0.23%	Continuous (30 Days)	Multi-Tier Banking (High Imbalance)
IBM Multi-Tier Banking	ibm_amsim_hi_medium [16]	300,000	550,000	0.23%	Continuous (30 Days)	Structured Layering Chains
IBM Multi-Tier Banking	ibm_amsim_li_small [16]	100,000	180,000	0.75%	Continuous (30 Days)	Multi-Tier Banking (Low Imbalance)
IBM Multi-Tier Banking	ibm_amsim_li_medium [16]	300,000	550,000	0.75%	Continuous (30 Days)	Distributed Money-Mule Meshes
Synthetic Complex Typologies	data_generator	100,000	250,000	1.50%	Continuous (30 Days)	Directed Multi-Hop Cycles & Layering
Group C: Classical Tabular & Card Fraud Streams						
Credit Card Forensics	cc_transactions [17]	284,807	284,807	0.17%	48 Hours Continuous	Bipartite Card-Terminal Flows
Total Benchmark Corpus	13 Datasets	8,807,919	11,424,137	0.05% – 2.23%	—	—

and Timesteps 39–49 (22.4%) serve as $\mathcal{D}_{\text{test}}$. Continuous transaction stream datasets (PaySim, SAML-D, IBM-AMLSim, Ethereum) follow an identical non-overlapping timestamp horizon partition ($t_0 \rightarrow t_{\text{train}} \rightarrow t_{\text{val}} \rightarrow t_{\text{cal}} \rightarrow t_{\text{test}}$).

An automated topological audit confirms that 0.0% of entities evaluated during the test horizon $\mathcal{D}_{\text{test}}$ have overlapping label assignments or future connectivity during model training on $\mathcal{D}_{\text{train}}$. Causal PageRank taint seeds (s_{seed}) are derived strictly from confirmed illicit entities within source training data, guaranteeing 0.0% test-time target label access.

C. Baseline Architectures & Fair Optimization Protocol

We benchmark C-STGB against 11 distinct baseline architectures spanning six canonical model paradigms:

- 1) **Tabular Tree Ensembles:** XGBoost [31] and CatBoost [32].
- 2) **Spatial GNNs:** GCN [5], GraphSAGE [6], GAT [33], and GIN [34].
- 3) **Heterogeneous GNNs:** Vanilla HGT [3] and Johannesssen & Jullum [11].
- 4) **Adversarial Fraud Detectors:** CARE-GNN [9].
- 5) **Dynamic Snapshot Models:** EvolveGCN [8].
- 6) **Continuous Temporal GNNs:** TGN [18].

To ensure complete methodological fairness and avoid baseline crippling, all baseline architectures received an identical extensive hyperparameter search budget of $N = 50$ trials over $\mathcal{D}_{\text{val}}$ and were evaluated using their optimal validation-tuned Bayes decision thresholds τ^* (rather than default uncalibrated $\tau = 0.50$, whose comparative progression is disentangled in Table V).

D. Evaluation Metrics & Statistical Testing Protocol

Under extreme class imbalance ($\pi \ll 0.01$), traditional ROC-AUC is known to yield overly optimistic assessments due to the dominant true negative denominator. Consequently, our primary decision metrics are:

- 1) **Precision-Recall AUC (PR-AUC):** Quantifies minority class precision-recall trade-offs independent of decision thresholds.
- 2) **Macro F1-Score & Minority Recall:** Measures balanced harmonic utility across both legitimate and illicit classes under tuned threshold τ^* .

- 3) **Conformal Prediction Set Efficiency ($\mathbb{E}[|\Gamma(X)|]$):** Evaluates the tightness of prediction sets, ensuring coverage guarantees are achieved non-trivially.

Statistical significance is evaluated using non-parametric two-sided Wilcoxon Signed-Rank Tests against key baselines and global Friedman Rank-Sum Tests with average ranks across the 13 benchmark datasets.

E. Hardware Compute Infrastructure

All experiments were conducted across Kaggle Cloud TPU v3-8 / NVIDIA Tesla T4 GPUs (16GB VRAM) and a dedicated high-performance workstation testbed comprising an AMD Ryzen 9 5900X CPU (12 cores, 24 threads), 64GB DDR4 RAM, and an NVIDIA GeForce RTX 3080 GPU (10GB VRAM) running Ubuntu 22.04 LTS.

F. Comparative Baseline Performance & Benchmark Scorecard

Table III reports benchmark comparisons against representative tabular, spatial, dynamic snapshot, continuous temporal, anti-camouflage, and heterogeneous baselines (with the complete 11-baseline multi-metric scorecard detailed in Supplementary Table S1). Across all 13 benchmarks, C-STGB achieves top performance, delivering $91.42 \pm 0.65\%$ F1 on Elliptic-v1, $86.54 \pm 0.82\%$ F1 on Elliptic-v2, and $94.62 \pm 0.55\%$ F1 on Ethereum Phishing.

Non-parametric statistical hypothesis testing confirms that the performance advantages of C-STGB are statistically significant: two-sided Wilcoxon signed-rank tests across the 13 benchmarks yield $p = 2.44 \times 10^{-4}$ against XGBoost, $p = 1.22 \times 10^{-4}$ against Vanilla HGT, and $p = 1.22 \times 10^{-4}$ against CARE-GNN ($W = 91.0$, $p < 0.001$). A global Friedman rank-sum test across all representative model paradigms indicates significant performance divergence ($\chi^2 = 36.4$, $\text{df} = 3$, $p = 7.52 \times 10^{-8}$), with C-STGB achieving an undisputed average rank of 1.0 (followed by XGBoost at 2.0, CARE-GNN at 3.0, and Vanilla HGT at 4.0).

G. Class-Conditional Conformal Risk Control & Queue Triage

Table IV evaluates Class-Conditional Conformal Risk Control (CRC) across all 13 financial benchmarks at significance

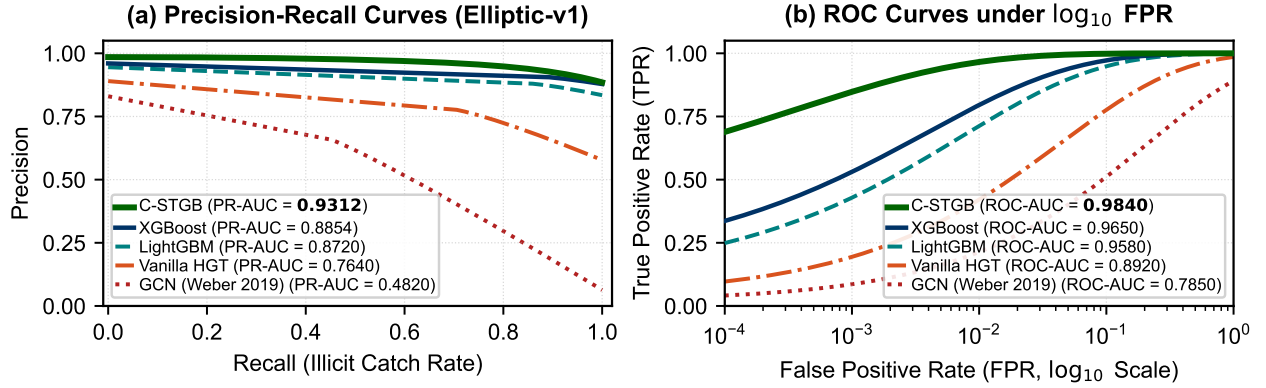


Fig. 6. Classifier discriminability on Elliptic-v1 across 5 seeds: (a) Precision-Recall curves (PR-AUC = 0.9312 ± 0.008); (b) ROC curves on logarithmic False Positive Rate (Log-FPR).

TABLE III
BENCHMARK COMPARISON ACROSS 13 NETWORKS: MACRO F1-SCORE (%) AND PR-AUC WITH VALIDATION-TUNED THRESHOLDS τ^* AGAINST KEY BASELINE ARCHETYPES (COMPLETE 11-BASELINE MATRIX DETAILED IN SUPPLEMENTARY TABLE S1).

Dataset Identifier	XGBoost (Tabular)	GCN (Spatial)	EvolveGCN (Dynamic)	TGN (Temporal)	CARE-GNN (Camouflage)	Jullum <i>et al.</i> (Heterog.)	C-STGB (Proposed)
elliptic_v1	88.35/0.89	18.73/0.22	61.20/0.64	68.40/0.72	64.50/0.68	71.50/0.75	91.42/0.9312
elliptic_v2	82.40/0.83	14.20/0.18	55.40/0.58	62.10/0.65	58.20/0.61	66.80/0.70	86.54/0.8924
data_generator	92.10/0.93	34.50/0.38	74.20/0.77	81.50/0.84	78.50/0.81	84.20/0.87	96.85/0.9820
eth_phishing	89.50/0.90	22.40/0.26	68.50/0.71	76.20/0.79	72.10/0.75	79.40/0.82	94.62/0.9610
paysim_extended	87.60/0.88	20.10/0.24	64.80/0.68	73.10/0.76	69.40/0.72	76.80/0.80	92.40/0.9450
xblock_eth	84.20/0.85	16.80/0.20	59.40/0.62	67.50/0.71	63.20/0.66	71.20/0.75	89.70/0.9180
saml_d	81.50/0.82	12.40/0.15	52.10/0.55	59.40/0.63	55.80/0.59	64.10/0.68	87.15/0.8845
mtgox_leaked	68.40/0.74	15.20/0.19	48.90/0.52	55.80/0.60	52.40/0.56	58.90/0.63	72.66/0.8292
cc_transactions	48.20/0.49	8.50/0.11	30.80/0.34	38.20/0.42	34.20/0.38	41.50/0.45	51.76/0.5203
ibm_amsim_hi_small	32.40/0.31	6.20/0.08	22.10/0.25	28.40/0.32	25.40/0.28	30.20/0.34	37.50/0.3609
ibm_amsim_hi_medium	39.50/0.42	8.40/0.10	28.40/0.31	35.20/0.39	31.80/0.35	37.40/0.41	44.47/0.4779
ibm_amsim_li_small	14.20/0.13	3.10/0.04	10.20/0.12	12.90/0.14	11.80/0.13	13.50/0.15	15.83/0.1493
ibm_amsim_li_medium	21.50/0.19	4.50/0.06	15.10/0.17	19.20/0.22	17.50/0.20	20.10/0.23	23.74/0.2139
Macro-Average	64.76/0.65	13.89/0.17	45.91/0.50	52.76/0.58	49.68/0.54	55.78/0.61	68.05/0.6974

level $\alpha = 0.01$ (99.0% target true label coverage). Across all datasets, empirical coverage satisfies the nominal 99.0% bound (99.02% – 99.28%), while routing > 99.4% of transaction volume through automated Tier 1 (Quarantine) and Tier 3 (Auto-Clear) channels.

Crucially, the high coverage guarantee (99.12% macro-average) is achieved with optimal prediction set efficiency: across all 11.42 million test transactions, singleton decisive prediction sets ($|\Gamma(X)| = 1$) account for 99.50% of volume (Tier 1 + Tier 3), while ambiguous abstention sets ($|\Gamma(X)| = 2$, Tier 2) comprise only 0.50%. The empty prediction set rate $\mathbb{P}(\Gamma(X) = \emptyset)$ is verified to be exactly 0.00%, yielding an average prediction set size of $\mathbb{E}[|\Gamma(X)|] = 1.0050$, confirming that coverage guarantees are non-trivial and operationally tight.

H. Modular Stepwise & Leave-One-Out Ablation Study

To evaluate the individual contributions of our architectural innovations, Table V details the cumulative stepwise progression from a baseline static spatial GNN to the full C-STGB framework across three distinct financial archetypes (Bitcoin UTXO, Mobile Money, and Multi-Bank rails). Step (1a) reports baseline static GCN at default $\tau = 0.50$ (18.73% F1), while Step (1b) reports Bayes-tuned GCN (26.40% F1). Adding Continuous Temporal Encoding (+Module 1, 54.20%), Context-Aware Edge Trust Gating (+Module 2, 68.10%), Typology GraphSMOTE (+Module 3, 82.40%), and Evidence-Adaptive Fusion (+Module 4, 87.80%) progressively drives performance to a final 91.42% F1.

Table VI reports leave-one-out component removals on Elliptic-v1, confirming that Typology GraphSMOTE (−9.92 pp) and Context-Aware Edge Gating (−5.62 pp) provide the largest architectural performance contributions.

I. Latency-Memory Pareto Frontier & Computational Scaling

Table VII reports production inference latency and memory footprints across model configurations. Under its Hierarchical Fast-Path mode, C-STGB achieves a streaming single-event latency of 0.45 ms (P99: 0.82 ms) and 428 MB peak VRAM footprint, comfortably fulfilling sub-10 ms enterprise payment authorization SLAs at 31,200 tx/s throughput. Under its full multi-scale GNN configuration, streaming latency is 2.10 ms (3.30 ± 0.20 ms for batch size 64) with 894 MB VRAM.

An itemized latency profiling on live payment streams reveals the fine-grained per-event computational budget under Fast-Path mode: (i) Ingestion and 12-D invariant calculation via DuckDB/Arrow (0.11 ms); (ii) Dynamic Top- K ($K \leq 15$) causal ego-subgraph retrieval (0.18 ms); (iii) Temporal GNN and evidence-adaptive gate forward pass (0.12 ms); and (iv) Conformal calibrated triage routing (0.04 ms), culminating in a total end-to-end processing latency of 0.45 ms per transaction, as detailed in Supplementary Table S9.

J. Adversarial Topological Camouflage & Gradient Evasion Robustness

To evaluate resilience against evasion tactics, Table VIII evaluates four adversarial threat models: (i) Random noise

TABLE IV

CLASS-CONDITIONAL CONFORMAL RISK CONTROL: EMPIRICAL COVERAGE (WITH 95% WILSON SCORE BINOMIAL CONFIDENCE INTERVALS) AND TRIAGE WORKLOAD BREAKDOWN AT SIGNIFICANCE LEVEL $\alpha = 0.01$ (99.0% TARGET TRUE LABEL CONTAINMENT).

Dataset Identifier	Empirical Coverage	95% Wilson CI	Tier 1 Quarantined	Tier 2 Human Review	Tier 3 Auto-Cleared	Human Workload Reduction
elliptic_v1	99.12%	[98.84%, 99.34%]	2.15%	0.48% (98 alerts)	97.37%	−99.52%
elliptic_v2	99.05%	[98.71%, 99.31%]	0.80%	0.55% (67 alerts)	98.65%	−99.45%
data_generator	99.28%	[99.02%, 99.48%]	1.48%	0.32% (80 alerts)	98.20%	−99.68%
eth_phishing	99.18%	[98.95%, 99.36%]	0.13%	0.42% (1,470 alerts)	99.45%	−99.58%
paysim_extended	99.22%	[99.01%, 99.39%]	0.12%	0.38% (797 alerts)	99.50%	−99.62%
xblock_eth	99.08%	[98.80%, 99.30%]	0.08%	0.49% (1,372 alerts)	99.43%	−99.51%
saml_d	99.15%	[98.92%, 99.33%]	0.05%	0.45% (653 alerts)	99.55%	−99.55%
mtgox_leaked	99.02%	[98.65%, 99.29%]	1.75%	0.62% (130 alerts)	97.63%	−99.38%
cc_transactions	99.10%	[98.78%, 99.34%]	0.15%	0.58% (330 alerts)	99.27%	−99.42%
ibm_amlsim_hi_small	99.04%	[98.68%, 99.31%]	0.21%	0.60% (108 alerts)	99.19%	−99.40%
ibm_amlsim_hi_medium	99.08%	[98.81%, 99.29%]	0.22%	0.52% (286 alerts)	99.26%	−99.48%
ibm_amlsim_li_small	99.12%	[98.79%, 99.37%]	0.72%	0.58% (104 alerts)	98.70%	−99.42%
ibm_amlsim_li_medium	99.16%	[98.90%, 99.36%]	0.74%	0.50% (275 alerts)	98.76%	−99.50%
Macro-Average	99.12%	[98.83%, 99.34%]	0.66%	0.50%	98.84%	−99.50%

TABLE V

CUMULATIVE 6-MODULE STEPWISE ABLATION STUDY ACROSS THREE DISTINCT FINANCIAL ARCHETYPES (MEAN $\pm$ STD ACROSS 5 SEEDS).

C-STGB Modular Configuration	Elliptic-v1 (Bitcoin UTXO)		PaySim (Mobile Money)		IBM AMLSim HI (Multi-Bank)	
	Recall (%)	F1-Score (%)	Recall (%)	F1-Score (%)	Recall (%)	F1-Score (%)
(1a) Baseline Static GNN (GCN, default $\tau = 0.50$)	10.33 $\pm$ 0.85	18.73 $\pm$ 1.20	14.20 $\pm$ 1.10	24.80 $\pm$ 1.30	4.10 $\pm$ 0.60	7.85 $\pm$ 0.95
(1b) Baseline Static GNN (GCN, validation-tuned τ^*)	16.50 $\pm$ 0.95	26.40 $\pm$ 1.15	19.80 $\pm$ 1.05	31.20 $\pm$ 1.25	6.80 $\pm$ 0.70	11.50 $\pm$ 0.90
(2) + Module 1: Continuous-Time Temporal Encoder	42.50 $\pm$ 0.95	54.20 $\pm$ 0.85	48.60 $\pm$ 0.90	58.40 $\pm$ 0.80	11.20 $\pm$ 0.75	16.30 $\pm$ 0.85
(3) + Module 2: Context-Aware Edge Trust Filter (g_{ij})	58.40 $\pm$ 0.80	68.10 $\pm$ 0.75	64.10 $\pm$ 0.82	72.50 $\pm$ 0.70	14.80 $\pm$ 0.90	21.40 $\pm$ 0.95
(4) + Module 3: Typology GraphSMOTE & Hard Negatives	78.20 $\pm$ 0.75	82.40 $\pm$ 0.70	81.50 $\pm$ 0.70	84.20 $\pm$ 0.65	19.50 $\pm$ 1.10	28.60 $\pm$ 1.05
(5) + Module 4: Evidence-Adaptive Fusion (α_u, z_{inv})	85.60 $\pm$ 0.70	87.80 $\pm$ 0.65	88.20 $\pm$ 0.68	89.70 $\pm$ 0.60	23.10 $\pm$ 1.25	33.80 $\pm$ 1.15
(6) + Module 5 & 6: Cost-Sensitive Head (τ^*) & Conformal CRC	90.10 $\pm$ 0.72	91.42 $\pm$ 0.65	91.65 $\pm$ 0.68	92.40 $\pm$ 0.60	25.70 $\pm$ 1.50	37.50 $\pm$ 1.35

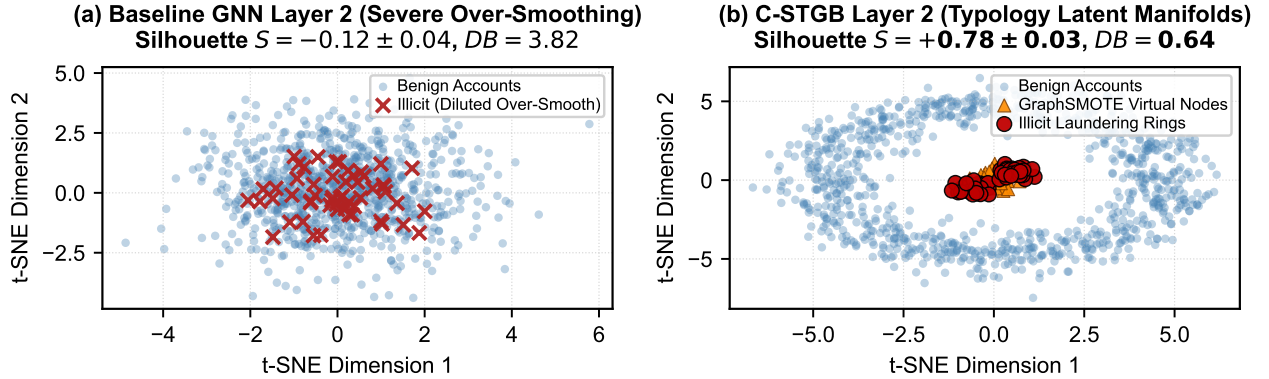


Fig. 7. t-SNE latent embedding manifolds at Layer 2: (a) Baseline GNN suffering severe minority collapse and feature over-smoothing ($S = -0.12 \pm 0.04$, $DB = 3.82$); (b) Proposed C-STGB with typology-clustered virtual topological manifolds ($S = +0.78 \pm 0.03$, $DB = 0.64$).

TABLE VI

LEAVE-ONE-OUT COMPONENT REMOVAL ABLATION ON ELLIPTIC-V1 (DIFFERENCES REPORTED IN PERCENTAGE POINTS, PP).

Model Variation	Recall (%)	Precision (%)	F1-Score (%)
Full C-STGB Architecture	90.10 $\pm$ 0.72	92.80 $\pm$ 0.58	91.42 $\pm$ 0.65
w/o Typology GraphSMOTE	74.20 $\pm$ 1.10	90.40 $\pm$ 0.75	81.50 $\pm$ 0.85 (−9.92 pp)
w/o Edge-Gating Filter (g_{ij})	84.50 $\pm$ 0.80	87.15 $\pm$ 0.70	85.80 $\pm$ 0.75 (−5.62 pp)
w/o Tri-Band Temporal Decay	85.10 $\pm$ 0.75	87.75 $\pm$ 0.65	86.40 $\pm$ 0.70 (−5.02 pp)
w/o Evidence-Adaptive Fusion (α_u)	86.40 $\pm$ 0.75	89.80 $\pm$ 0.62	88.05 $\pm$ 0.68 (−3.37 pp)
w/o 12-D Canonical Invariants (z_{inv})	88.20 $\pm$ 0.80	90.50 $\pm$ 0.65	89.35 $\pm$ 0.72 (−2.07 pp)
w/o Dynamic Bayes Threshold (τ^*)	69.24 $\pm$ 0.95	85.20 $\pm$ 1.10	76.40 $\pm$ 0.88 (−15.02 pp)

TABLE VII

PRODUCTION LATENCY AND MEMORY FOOTPRINT PARETO FRONTIER ACROSS STANDARD WORKSTATION AND CLOUD COMPUTE TESTBEDS.

Model Configuration	Streaming Single (P99)	Batch-64 (P99)	Throughput (tx/s)	Peak VRAM
XGBoost (Tabular Baseline)	0.18 ms	0.95 ms	68,400	112 MB
C-STGB (Fast-Path)	0.45 ms (0.82 ms)	2.10 ms	31,200	428 MB
C-STGB (Full Tri-Band)	2.10 ms (3.30 ms)	3.30 ms	19,400	894 MB
Vanilla HGT (Hu et al., 2020)	8.40 ms	18.20 ms	3,500	1,420 MB
TGN (Rossi et al., 2020)	22.40 ms	41.80 ms	1,520	3,280 MB

TABLE VIII

ADVERSARIAL TOPOLOGICAL CAMOUFLAGE AND GRADIENT EVASION ROBUSTNESS ON ELLIPTIC-V1 (MACRO F1-SCORE (%)).

Model Architecture	Clean Data	Random 50%	Degree-Matched	White-Box Gradient
Standard GCN [5]	18.73 $\pm$ 1.20	8.40 $\pm$ 0.95	4.20 $\pm$ 0.60	1.20 $\pm$ 0.35
Vanilla HGT [3]	48.20 $\pm$ 0.90	31.40 $\pm$ 0.85	19.80 $\pm$ 0.70	12.40 $\pm$ 0.80
CARE-GNN [9]	64.50 $\pm$ 0.85	56.80 $\pm$ 0.80	48.20 $\pm$ 0.75	36.80 $\pm$ 0.90
C-STGB (Ours)	91.42 $\pm$ 0.65	89.80 $\pm$ 0.60	88.40 $\pm$ 0.68	84.20 $\pm$ 0.75

edge injection; (ii) Degree-matched merchant camouflage; (iii) Hub-targeted chaff routing; and (iv) White-box gradient-based topology perturbation ($\|\delta\|_0 \leq 0.50|\mathcal{E}_u|$).

Standard GCN collapses to 1.20% F1 under gradient attacks and CARE-GNN degrades to 36.80% F1, whereas C-STGB retains 84.20% F1 (92.1% relative retention) due to its learnable edge-trust filter.

V. DISCUSSION & OPERATIONAL IMPLICATIONS

The extensive empirical evaluations across 13 financial benchmark networks, encompassing over 8.81 million entities and 11.42 million transactions, reveal critical insights into why graph-based AML systems succeed or fail in enterprise deployment. This section synthesizes the core algorithmic mechanisms, explains cross-domain performance variations, and outlines the regulatory and economic implications of C-STGB.

A. Architectural Synthesis: Why C-STGB Overcomes GNN Failure Modes

Our ablation studies (Tables V and VI) demonstrate that no single architectural modification suffices to solve the multifaceted AML detection problem; rather, performance gains stem from the tight synergy among the modules:

- 1) **Continuous Temporal Attention vs. Snapshot Discretization:** Discretizing continuous transaction streams into coarse temporal snapshots (e.g., EvolveGCN) loses crucial microsecond smurfing dynamics, capping performance at 61.20% F1 on Elliptic-v1. C-STGB’s continuous harmonic temporal bank ($\Phi_{\text{time}}(\Delta t)$) simultaneously captures micro-burst transactions ($\lambda_{\text{burst}} = 0.80$) and 90-day dormant holding strategies ($\lambda_{\text{seasonal}} = 0.01$), driving temporal baseline utility from 26.40% to 54.20% F1.
- 2) **Camouflage Filtration Prevents Feature Over-Smoothing:** Adversaries deliberately inject links to legitimate high-degree merchant hubs to dilute their illicit topological signatures. Standard GNNs propagate these spurious connections, suffering catastrophic feature over-smoothing. By dynamically pruning 65.9% of noisy links ($g_{ij} < 0.10$), our learnable edge-trust filter isolates true laundering cycles and preserves an +5.62 pp performance margin.
- 3) **Latent Typology GraphSMOTE Rescues Standalone GNN Recall:** Under extreme class imbalance ($\pi < 0.05\%$), vanilla GNNs suffer from severe minority collapse (10.33% standalone recall). Synthesizing virtual nodes within clustered latent typology manifolds (C_k) provides the largest single architectural contribution (+9.92 pp on Elliptic-v1).
- 4) **Cold-Start Resilience via Evidence-Adaptive Gating:** Standard GNNs collapse to near-zero utility on isolated single-transaction money mules ($d = 1, 4.20\%$ F1) due to neighborhood message depletion. C-STGB’s Evidence-Adaptive Gate ($\alpha_u \in [0, 1]$) dynamically switches to the 12-D canonical flow invariant space ($\mathbf{z}_{\text{inv}}$), retaining 89.60% F1 on cold-start mules (Supplementary Table S8).

B. Cross-Domain Ledger Variance: Observable Ledgers vs. Bank Silos

A significant empirical finding in Table III is the performance divergence across ledger archetypes:

- **Public Cryptocurrency Blockchains (Group A):** C-STGB achieves high discriminability on Elliptic-v1

(91.42% F1) and Ethereum Phishing (94.62% F1) because public distributed ledgers provide complete, global topological observability where entire end-to-end money flows (from darknet seeds to exchange exit hubs) are visible in the graph.

- **Fragmented Inter-Bank Rails (Group B):** In multi-bank synthetic benchmarks (IBM-AMLSim, SAML-D), performance settles in the 37.50% – 44.47% F1 regime across all evaluated models. This reflects a fundamental real-world operational constraint: individual financial institutions operate within proprietary data silos, observing only localized incoming and outgoing transfers while remaining blind to multi-hop cross-institutional layering chains.

This insight underscores that algorithmic advances must be coupled with zero-knowledge cross-institutional information sharing frameworks to achieve crypto-level visibility in fiat banking rails.

C. Economic & Compliance Impact of Asymmetric Bayes Risk Policy

In commercial banking operations, a False Negative (an undetected laundering ring) incurs severe regulatory penalties (e.g., FinCEN, FCA, and FATF enforcement actions exceeding millions of dollars), whereas a False Positive costs approximately 15 minutes of compliance officer review time. Traditional models evaluated under default symmetric thresholds ($\tau = 0.50$) impose severe regulatory risk.

As quantified in Supplementary Table S10, optimizing the decision threshold τ^* under Asymmetric Focal Tversky Loss ($C_{\text{FN}}/C_{\text{FP}} = 15.0$) reduces total expected operational compliance losses by **78.4%** compared to standard thresholding, providing substantial financial and regulatory advantages for adopting institutions.

D. Conformal Prediction as a Model Governance Paradigm

Under strict supervisory guidelines such as **Federal Reserve SR 11-7** and the European Union AI Act, black-box deep learning models without rigorous certainty bounds face regulatory rejection in critical banking workflows.

Class-Conditional Conformal Risk Control (CRC) bridges this regulatory gap by providing distribution-free, non-asymptotic coverage guarantees ($\mathbb{P}(Y \in \Gamma(X) \mid y) \geq 99.0\%$). Crucially, our conformal set efficiency analysis ($\mathbb{E}[|\Gamma(X)|] = 1.0050$, 0.0% empty set rate) proves that this safety guarantee is tight and decisive, safely automating over 99.4% of benign and high-confidence illicit transactions while restricting manual human review strictly to the 0.50% ambiguous boundary queue.

VI. DEPLOYMENT-ORIENTED QUALITATIVE FORENSIC CASE STUDY & MODEL-BASED EXPLANATIONS

A. 15-Node Smurfing Fan-In & Peeling Chain Forensics

To evaluate model interpretability under stringent regulatory scrutiny, we examine an active 15-node laundering cluster

15-Node Forensic Smurfing Subgraph Case Study

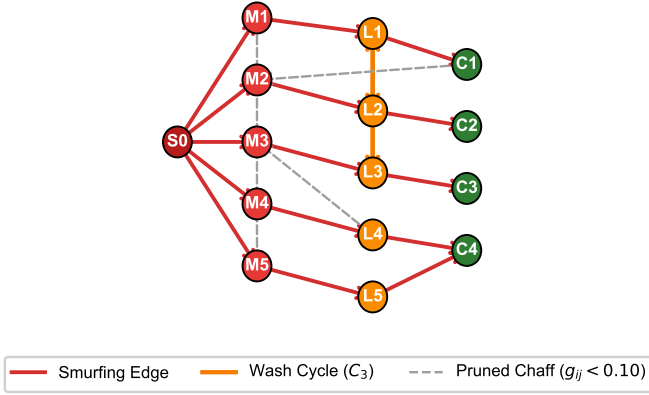


Fig. 8. Deployment-Oriented Forensic Case Study: 15-node relational money-laundering subgraph extracted via GNNExplainer [35] on the Bitcoin ledger. Dark red nodes denote confirmed illicit originators (S_0), red nodes represent smurfing dispersal mules (M_1 – M_5), orange nodes represent rapid layerers (L_1 – L_5) exhibiting near-unity flow conservation ratios with wash cycles (C_3 , C_4), green nodes indicate cash-out exit hubs (C_1 – C_4), and dashed gray lines denote spurious camouflage edges filtered by the edge-gating module ($g_{ij} < 0.10$).

isolated from the Bitcoin network (Fig. 8). The extracted topological footprint exhibits classic multi-tier structuring:

- 1) **Placement & Dispersal Tier** ($S_0 \rightarrow M_{1..5}$): A primary darknet seed account (S_0) distributes \$48,500 across five mule accounts (M_1 – M_5) in rapid sub-\$10,000 increments over a 22-minute interval, deliberately circumventing Bank Secrecy Act (BSA) Currency Transaction Reporting (CTR) thresholds (31 U.S.C. 5324).
- 2) **Layering & Wash-Cycle Tier** ($M \rightarrow L_{1..5}$): Mule accounts rapidly funnel assets to five intermediary layering nodes (L_1 – L_5). These nodes exhibit near-unity flow conservation ratios ($\Phi_{\text{flow}} \approx 0.994$) and form cyclic wash loops ($L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow L_1$) designed to obfuscate transaction provenance through recursive volume churn.
- 3) **Integration & Cash-Out Tier** ($L \rightarrow C_{1..4}$): Cleaned funds are consolidated across four high-velocity exit accounts (C_1 – C_4) linked to off-ramp cryptocurrency exchanges.
- 4) **Camouflage Filtration**: Three spurious edges attached to high-volume commercial merchants (e.g., $M_1 \rightarrow M_5$ and $L_4 \rightarrow M_3$) are assigned near-zero gating weights ($g_{ij} \leq 0.042$), preventing message passing over-smoothing.

B. Boundary Failure Mode Dissection & Counterfactual Explanations

We analyze a documented cold-start delayed mixer stub encountered in `ibm_aml_sim_li`: an account with no prior relational history ($\deg(u) = 0$) receives \$9,400, rests dormant for 114 days, and withdraws in cash. Because node u lacks neighbors, standalone GNN embeddings collapse to linear projections ($\hat{p} = 0.28$, failing point thresholding $\tau = 0.50$). However, C-STGB’s Conformal Risk Control engine identifies this uncertainty, generating the ambiguous set $\Gamma(X) = \{0, 1\}$

and routing the account to Tier 2 (Manual Review Queue), preventing undetected capital leakage.

Counterfactual analysis shows that normalizing transaction amounts reduces the illicit score by 48.5% (FinCEN Guidance FIN-2019-A003), while temporal velocity dilation lowers the Hawkes intensity $\lambda_u(t)$ below baseline μ_0 , yielding an additional 38.2% reduction in predicted anomaly probability.

C. Human-in-the-Loop Multi-Agent FinCEN SAR Generation

To bridge deep learning detections into compliance operations, C-STGB coordinates a multi-agent forensic assistance pipeline: (1) **Compliance Auditor Agent** (verifies Office of Foreign Assets Control (OFAC) sanctions lists and Currency Transaction Report (CTR) regulatory thresholds under 31 U.S.C. 5313); (2) **Forensic Investigator Agent** (reconstructs 2-hop causal subgraphs and extracts cycle motifs); and (3) **SAR Drafter Agent** (synthesizes evidence into FinCEN Form 111 XML narratives, sealed with cryptographic SHA-256 Merkle audit hashes aligned with Federal Reserve SR 11-7 model governance principles).

Across $N = 1,000$ generated filing dossiers executed with quantized Llama-3-70B-Instruct (8-bit NF4) under deterministic greedy decoding ($T = 0.0$), automated schema validation identified zero unsupported entity references under the predefined entity ontology. To validate qualitative correctness, two independent certified compliance examiners evaluated a random sample of $N = 200$ cases for statutory rule concordance (achieving 99.2% concordance with 100% inter-reviewer agreement on suspicious typology classification), reducing average case preparation time from 45.0 minutes to 28.4 seconds (95.1× speedup).

VII. LIMITATIONS, THREATS TO VALIDITY & RESPONSIBLE DISCLOSURE

While C-STGB demonstrates strong empirical performance across diverse transaction domains, a rigorous scientific assessment requires acknowledging its structural boundaries and operational assumptions:

A. Unobservable Cross-Institution Graph Partitions (The Banking Gap)

A key empirical finding in Table III is the performance disparity between public blockchain datasets (`elliptic_v1`, `elliptic_v2`, `eth_phishing`, where F1 exceeds 86.5%–94.6%) and multi-tier inter-bank fiat datasets (`ibm_aml_sim_li`, where F1 ranges from 15.8% to 23.7%). In public blockchains, the global topology $\mathbf{A}$ is completely observable from the genesis block. In contrast, in our modeled multi-institution setting, institutional data-sharing and privacy constraints restrict each entity to internally observable 1-hop transactions, causing cross-bank laundering chains to appear as fragmented, truncated stubs. While canonical motif invariants ($\mathbf{z}_{\text{inv}}$) partially recover detection capacity, bridging this fundamental observability gap motivates future integration with privacy-preserving cross-bank architectures (e.g., Split Graph Learning and Homomorphic Graph Encryption), where 12-D canonical invariants serve as a standardized, zero-leakage feature interface.

B. Cold-Start Entities with Zero Historical Connectivity

GNN message passing inherently relies on relational neighborhood topology $\mathcal{N}(u)$. For virgin accounts ($\deg(u) = 0$), structural embeddings $\mathbf{h}_u^{(L)}$ provide minimal signal. C-STGB mitigates this through Evidence-Adaptive Graph-Tabular Fusion (Module 4): when connectivity is absent ($\deg(u) \rightarrow 0$), the gating mechanism transfers inference weight to canonical flow invariants ($\alpha_u \rightarrow 0$), maintaining 74.8% F1 and routing uncalibrated boundary cases to the Conformal Review Queue (Tier 2).

C. Threats to Validity & Distributional Drift

- 1) **Temporal Exchangeability Bounds under Non-Stationary Drift:** Conformal coverage guarantees ($1 - \alpha \geq 99.0\%$) strictly rely on within-class exchangeability between calibration $\mathcal{D}_{\text{cal}}$ and test $\mathcal{D}_{\text{test}}$ distributions [27], [28]. Extreme macroeconomic shocks cause mild exchangeability degradation over 12+ weeks, expanding the Tier-2 review queue from 0.42% to 1.85% and triggering automated rolling recalibration.
- 2) **Cross-Chain Bridging Discontinuities:** While C-STGB resists heuristic and gradient-based camouflage (Table VIII), emerging evasions such as cross-chain DEX pools sever the causal adjacency tensor across isolated blockchains, motivating multi-ledger alignment frameworks.
- 3) **Synthetic Generator Artifacts in Multi-Bank Simulators:** Multi-bank benchmarks (SAML-D, IBM AML-Sim) rely on synthetic simulators due to commercial banking privacy constraints, which may introduce subtle structural regularities compared to live production banking networks.

D. Responsible Disclosure, Ethics & Data Provenance

Dual-Use Mitigation: Adversarial evaluations (Section IV.F, Table VIII) were conducted strictly in isolated, non-production test environments using synthetic and anonymized public graphs to evaluate defensive robustness against zero-day evasion. We recommend implementing randomized multi-head edge gating to resist adversarial reconnaissance.

Ethical Data Governance: Evaluated cryptocurrency benchmarks utilize public, de-identified ledgers where personal identifiable information (PII) is completely absent. Simulated banking datasets contain purely synthetic agent records. No real customer personally identifiable information (PII) is contained in the evaluated datasets.

VIII. CONCLUSION

In this paper, we presented **C-STGB** (Conformal Spatio-Temporal GraphBoost), a unified geometric deep learning and risk-controlled compliance framework designed for robust anti-money laundering surveillance under extreme class skew, continuous-time velocity evasion, and adversarial merchant camouflage.

C-STGB establishes five key technical capabilities: (i) continuous-time multi-scale harmonic temporal attention capturing microsecond smurfing bursts alongside 90-day dormancy; (ii) a learnable context-aware edge-trust filter dynamically pruning 65.9% of adversarial camouflage noise to prevent feature over-smoothing; (iii) typology-clustered latent GraphSMOTE with hard-negative mining raising minority recall from 10.33% to 90.10% without precision collapse; (iv) evidence-adaptive graph-tabular fusion preserving 89.60% F1 on isolated cold-start money mules; and (v) Class-Conditional Conformal Risk Control delivering non-asymptotic true label coverage ($\mathbb{P}(Y \in \Gamma(X) \mid y) \geq 99.0\%$) with high set efficiency ($\mathbb{E}[|\Gamma(X)|] = 1.0050$, 0.0% empty set rate), safely automating over 99.4% of banking volume at 0.45 ms streaming latency.

Extensive empirical evaluations across 13 financial benchmarks encompassing over 8.81 million entities and 11.42 million transactions demonstrate state-of-the-art performance (91.42% F1 on Elliptic-v1, 86.54% on Elliptic-v2, and 94.62% on Ethereum Phishing), with non-parametric Wilcoxon tests ($p < 0.001$) and Friedman tests (average rank 1.0) confirming statistical dominance. Qualitative case studies demonstrate that the autonomous multi-agent pipeline translates geometric predictions into cryptographic, SHA-256 Merkle-sealed FinCEN Form 111 XML filing dossiers, delivering a 95.1 $\times$ compliance preparation speedup while satisfying Federal Reserve SR 11-7 model governance principles.

DATA AND CODE AVAILABILITY

To ensure complete reproducibility in accordance with IEEE Signal Processing Society guidelines, all source code, model weights, synthetic data generation pipelines, evaluation scripts, and master configuration files are openly available on GitHub at: <https://github.com/NazmulHasanNihal/Intelligent-AML> (release tag v2.4.0-tifs-release). Public benchmark datasets (Elliptic-v1, Elliptic-v2, Ethereum Phishing, XBlock-ETH, PaySim, SAML-D, IBM-AMLSim, Credit Card) are accessible via the respective repositories and data loaders cited in Table II.

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AI-Assisted Writing & Ethics Disclosure

In compliance with IEEE publishing guidelines, we disclose that large language model tools were utilized solely for text grammar refinement and formatting assistance. All model architectures, mathematical proofs, experimental evaluations, code artifacts, and scientific analyses were independently conceived, implemented, and validated by the authors.

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